VAPI Track-1 Prerequisite Research: DualSense Edge Sensor-Stack Characterization

Document ID: VAPI-SENSOR-CHAR-001 **Status:** Canonical prerequisite anchor for v2 L4 feature architecture **Companion document:** BT calibration prerequisite research (BT-CALIB-LESSON-001 family) **Hardware under analysis:** Sony DualSense Edge wireless controller (CFI-ZCP1), USB VID 0x054C / PID 0x0DF2 (Edge), 0x0CE6 (standard DualSense)

TL;DR

- **Adaptive trigger force-curve telemetry → PRIMARY DISCRIMINATOR.** Vendor surface verified through `hid-playstation.c`, PSDevWiki, and the SensePost reverse-engineering work; force telemetry is a continuous 8-bit-per-trigger axis at the HID input rate (≈1 kHz on the DualSense Edge over USB by default). Literature-anchored inter-subject separation is comparable to or exceeding the Cross-button fingerprint: continuous-force biometrics achieve ~1% EER under high-fidelity capture (Saevanee 2009; piezo-touch NN ≈1.09% EER), 6.9–12.9% EER on Android pressure-augmented keystroke (Antal 2015; Chang 2012), and 76.8–89.6% identification on driving-pedal corpora (Wakita 2006; Miyajima 2007). Adaptive triggers additionally support a programmable-resistance challenge-response channel structurally analogous to the L6 haptic challenge. PCGamingWiki ResearchGate
- **Touchpad → CO-SIGNAL.** Touchalytics-class EERs (0–4%) come from swipe-velocity / contact-area features that the DualSense touchpad's 12-bit X/Y, 2-point capacitive encoding can partially support — but the touchpad is rarely engaged during competitive gameplay, so data-volume per session is the binding constraint, not separability. Additionally, the highest-mutual-information Touchalytics features (contact area, pressure) are not exposed by the DualSense HID stream.
- **Microphone array → CONDITIONALLY DROPPED (privacy-falsified at default scope).** The falsification position lands against the surface as a default-on channel in US/EU jurisdictions: any feature pipeline that captures speech-band audio triggers BIPA voiceprint liability ($1,000–$5,000 per scan) and the EU AI Act's biometric-categorization rules. A narrow non-speech sub-band may survive but is feasibility-research-only and not on the v2 L4 critical path. K&L Gates
- **Lightbar → CO-SIGNAL (third-party-witness challenge-response only).** Useful as a tournament-station optical witness path comparable in role to the LAN-tower BlueZ witness, but adds no per-player discriminative information by itself.
- **Battery drain multi-component signature → ADVISORY ONLY.** Battery telemetry is a 4-bit (11-bucket, ~10%-per-step) value updated at multi-minute cadence — too coarse to differentiate held-and-played vs. desk-resting controllers within per-match timescales.

- **Stick analog noise floor → CO-SIGNAL; Hall-effect stick fingerprinting → ADVISORY ONLY, session-bound presence only.** Standard HID exposes 8-bit per-axis on DualSense and DualSense Edge; this is at the resolution floor for resolving 8–12 Hz physiological hand tremor. The same-controller-population separability constraint that landed against L8 BT applies here verbatim: N≈50 identical Edges in a tournament hall will sit closer in Hall-stick response-curve feature space than a heterogeneous population. **The Hall-effect surface is marked session-bound presence attestation only, not cross-session controller identity, until a same-model separability study is run.** American Academy of Fami...

The defended position lands: **adaptive trigger force-curve is the leading L4 v2 candidate.** The defended position on novelty also lands: ARES 2024's audit set (BattlEye / EAC / FACEIT / Vanguard) plus 2025–2026 RICOCHET updates focuses entirely on host-OS-side rootkit-class instrumentation (kernel drivers, hypervisor checks, TPM 2.0 / Secure Boot attestation in RICOCHET Season 02). None of them collect adaptive-trigger force curves, controller-side touchpad capacitance, controller-mic acoustic fingerprints, lightbar optical proofs, or controller battery drain. **VAPI is structurally first in this surface set.** The Moonlight Call of Duty

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Verification-Block Discipline (BT-CALIB-LESSON-001 Application)

Every feature claim against a sensor surface below carries three verification classes:

1. **Vendor-spec confirmation** (DualSense reverse-engineering corpus): nondebug DualSense Explorer, `dualsensectl`, the SensePost reverse-engineering effort (`sensepost.com/blog/2020/dualsense-reverse-engineering/`), `psdevwiki.com/ps5/DualSense`, the Game Controller Collective Wiki (`controllers.fandom.com`), `pydualsense`, `dualsense-controller-python` (yesbotics).
2. **Open-source driver validation**: Linux mainline `drivers/hid/hid-playstation.c`, originally authored by Roderick Colenbrander (Sony/Gaikai), mainlined Linux 5.12 (April 2021); ALSA usb-audio Sony DualSense quirk patch series (Cristian Ciocaltea, Collabora, May 2025).
3. **Independent literature anchor** for the human-presence / biometric claim on that specific surface.

If any one of the three classes is missing, the surface is marked "unverified" and feature derivation is halted for that surface in v2 L4.

Surface 1 — Adaptive Trigger Force-Curve Telemetry (PRIMARY, 35% weight)

Vendor-spec confirmation

The DualSense L2/R2 triggers are mechatronic actuators with internal voice-coil motors (Sony codename "Walther" for the trigger assembly per the PSDevWiki board-component listing) capable of programmable resistance, plus continuous analog position/force telemetry.

HID input report fields (USB report ID 0x01, 64-byte report; BT report ID 0x31, 78-byte report):

- Byte 5: `Generic Desktop / Rx` — L2 trigger axis, 8-bit unsigned, neutral 0x00 → fully depressed 0xFF (per nondebug/dualsense parsed HID descriptor and Game Controller Collective Wiki `BasicGetStateData.TriggerLeft`). GitHub
- Byte 6: `Generic Desktop / Ry` — R2 trigger axis, 8-bit unsigned, same range.
- Trigger telemetry is reported every input report; on the **DualSense Edge over USB the default polling rate is 1000 Hz** (PCGamingWiki Controller:DualSense Edge entry, verified by user Yuuyatails 2024-08-27; DSX developer statement on Steam Community), versus ~250 Hz on the standard DualSense over USB and ~133–250 Hz over BT depending on host stack and firmware. PCGamingWiki + 2
- **The host can read back the user's applied force while a trigger effect is active** — DualSense-Windows (Ohjurot) explicitly documents *"Controlling the adaptive triggers (3 Types of effects) and reading back the users force while active."* `pydualsense` and `dualsense-controller-python` expose the same surface. GitHub

HID output report fields (USB report ID 0x02, 47-byte report; BT report ID 0x31, 78-byte report) carry trigger effect commands per Game Controller Collective Wiki `FFB Trigger Effect Factories`: `Off`, `Feedback`, `Weapon`, `Vibration`, `Slope`, `MultiplePositionFeedback`, etc. Each effect parameterizes resistance start position (0–9 in trigger travel zones), end position, strength (1–8), and frequency for vibration mode. **This is a programmable-resistance challenge channel.** GitHub

Open-source driver validation: `drivers/hid/hid-playstation.c` (Colenbrander, Linux 5.12+) parses both USB report 0x01 and BT report 0x31. The kernel structure `dualsense_input_report_common` carries `rx` (L2) and `ry` (R2) byte fields. Output report struct `dualsense_output_report_common` carries `right_trigger_motor_mode` and `left_trigger_motor_mode` plus 10-byte parameter blocks per trigger.

SensePost reverse-engineering effort (Stutman, November 2020) confirms the report-ID/report-size split between USB and BT modes and audio-class endpoint enumeration relevant to the parallel mic-array surface.

Per-firmware-version differences: firmware 0630 (2025-09-17) added multi-device BT pairing without changing trigger input encoding (PCGamingWiki). The Edge revision (CFI-ZCP1, USB PID 0x0DF2) preserves the 8-bit trigger encoding but ships with **Hall-effect trigger sensors** rather than the original DualSense's potentiometer-based trigger position sensing (gamepadtest.app PS5 controller tester documentation). This means trigger position telemetry on the Edge is drift-resistant and its reported value is closer to true mechanical position than on a worn standard DualSense. PCGamingWiki

Literature anchor (continuous-force biometrics)

Driving-simulator pedal-force biometrics:

- Wakita et al., "Driver Identification Using Driving Behavior Signals," *IEICE Trans. Inf. Syst.* E89-D (2006): GMM on accelerator + brake + velocity — **81% identification on N=12 simulator drivers, 73% on N=30 real-vehicle drivers.** arxiv Transactions Online
- Miyajima et al., "Driver Modeling Based on Driving Behavior and Its Evaluation in Driver Identification," *Proc. IEEE* (2007): cepstral features on gas + brake pedal pressure — **76.8% identification on N=276 drivers (real city driving), 89.6% in simulator.** arXiv
- Enev, Takakuwa, Koscher, Kohno, "Automobile Driver Fingerprinting," *PoPETs* 2016: random forest on CAN-bus pedal/steering — **claimed ~100% identification on N=15 drivers from brake-pedal alone.** *Replication caveat: Gahr et al., IEEE IV 2018, did not reproduce the perfect-ID claim on broader naturalistic data.* arxiv
- Fugiglando et al., "Driving Behavior Analysis through CAN Bus Data in an Uncontrolled Environment," *IEEE TITS* 20(3), 2019: **76% accuracy 2-driver, ~50% accuracy 5-driver on N=64 drivers in uncontrolled settings** — separability degrades from controlled to in-the-wild conditions. nih

MIDI key-velocity / piano-touch biometrics:

- Van Vugt, Jabusch, Altenmüller, "Individuality That is Unheard of: Systematic Temporal Deviations in Scale Playing Leave an Inaudible Pianistic Fingerprint," *Frontiers in Psychology* 4:134 (2013): minimum-Euclidean-distance on note-onset timing-deviation traces — **100% recognition on N=8 within-session, 83–100% on N=18 across muted/normal piano, above chance at 20-month follow-up.** Inter-class Euclidean distance ~7 ms vs. intra-class ~3 ms → separation ratio ≈ 2.3. ResearchGate
- Stamatatos & Widmer, "Automatic Identification of Music Performers with Learning Ensembles," *Artificial Intelligence* 165(1), 2005: **~70% identification on N=22 pianists (Bösendorfer SE) under inter-piece conditions** vs. ~5% chance.
- Bernays & Traube, "Investigating pianists' individuality...," *Frontiers in Psychology* 5:157 (2014): qualitative anchor — significant inter-pianist separation on touch/dynamics features.

Press-and-hold / pressure-dwell biometrics (closest analog to a single trigger-pull):

- Saevanee & Bhatarakosol, "Authenticating User Using Keystroke Dynamics and Finger Pressure," IEEE CCNC 2009: k-NN on touchpad pressure — **EER ≈ 1% on N=10 users in single-session lab conditions.** Treat as upper bound of separability. Academia.edu
- Antal, Szabó, László, "Keystroke Dynamics on Android Platform," *Procedia Technology* 19, 2015: random forest on Nexus-7/LG-Optimus password keystroke with timing + pressure + finger-area — **best EER 12.9% on N=42 users multi-session** (down from 15.3% timing-only). ResearchGate
- Chang, Tsai, Lin, *J. Systems & Software* (2012): **EER 12.2% timing-only → 6.9% with pressure** on N=100 + 20 imposters. ResearchGate
- Piezo-NN keystroke biometric in IoT setting: **EER 1.09%** with neural-network fusion of force + timing. ResearchGate

Inter-subject separation-ratio estimate for L2/R2 force-curve under VAPI conditions: Realistic field expectation for a held-trigger-pull-only feature set in the same-platform, multi-player setting is **EER 5–15% per pull**, with separability strongly dependent on whether the pull is during a high-information event (firing weapon, accelerating) or a passive low-amplitude event. This brackets the Cross-button transition-timing fingerprint (NCAA Football 26 corpus) and exceeds it in lab-best conditions (1% EER) but degrades in the field (per Fugiglando 2019, 5-driver uncontrolled accuracy ≈ 50%). On the protocol's separation-ratio framing, force-curve pull events **clear the 0.5 floor decisively.**

Separation-relevant features (literature-supported): peak force amplitude, ramp slope (dF/dt during onset), plateau hold time at fully-depressed position, release velocity, force-time integral (F·dt), pull duration, count and timing of micro-modulations during sustained pull (related to physiological tremor at 8–12 Hz, Lakie et al. 2012). nih

Adversarial spoof cost

Cronus Zen, XIM Matrix, and reWASD synthesize button/stick/trigger values via macro/script engine but do **not** synthesize *force-vs-time curves* with realistic biomechanical structure. The Cronus Zen "Expert Controller GamePack" exposes "Right/Left Trigger Deadzone Min/Max" and "Custom Analog Stick Curve" with 10 configurable points — this is parametric output remapping, not biomechanical-trace synthesis. RICOCHET Anti-Cheat Season 02 (https://www.callofduty.com/blog/2026/02/call-of-duty-black-ops-7-ricochet-anti-cheat-season-02, launched February 5, 2026) explicitly targets "machine-modified inputs" by analyzing input *behavior* (timing, consistency, response patterns) — confirming the threat model and confirming that no public translator-class device today produces convincing force-curve fakes. Producing a realistic continuous-force replay would require either (a) physical actuation of the trigger by a robotic finger driving a real DualSense, or (b) a deeper firmware-level intercept that synthesizes byte-level trigger telemetry mid-stream — neither is publicly demonstrated against this signal class. **Estimated spoof cost: very high.** Hardware: a robotic finger on a stand or a custom MITM-firmware bridge in the controller's USB/BT path. Skill: embedded firmware reverse-engineering plus biomechanical curve synthesis. Access: physical custody of a controller and per-target enrollment data. There are no published demonstrations of force-curve replay attacks against any continuous-force biometric system in the surveyed literature. Cronusmax + 2

Adaptive trigger as a challenge-response channel

The host can issue a programmed resistance pattern via output report — e.g., a `Slope` effect with start/end positions and start/end strengths, or a `Vibration` effect with specific frequency — and observe the player's force-vs-time response. This is structurally identical to the L6 haptic challenge channel currently shipping in the protocol, with the additional advantage that the response signal (player force) is bounded by physiological force-velocity envelopes that are not trivially synthesizable by current translator-class hardware. **Recommendation: implement adaptive-trigger challenge-response as L4-coupled-to-L6 in v2.**

Recommended L4 inclusion tier: PRIMARY DISCRIMINATOR

Surface 2 — Touchpad Capacitive Signal (CO-SIGNAL, 20% weight)

Vendor-spec confirmation

The DualSense touchpad is a 2-point capacitive surface with mechanical click. Per nondebug/dualsense parsed HID descriptor and the Game Controller Collective Wiki `TouchFingerData` struct:

- Each finger reports `Index : 7-bit, NotTouching : 1-bit, FingerX : 12-bit, FingerY : 12-bit`. Fandom
- Two fingers per touch packet, plus an 8-bit `Timestamp` for the touch frame.
- **No native pressure proxy or contact-area field is exposed** — this is a position-only encoding with binary contact state. Frank et al.'s Touchalytics features include "area covered by fingertip" and "fingertip pressure," neither of which is reported by the DualSense touchpad's HID stream. Academia.edu
- Cycling '74 forum thread confirms the X axis encoding is a 12-bit value packed into a 24-bit triplet shared with the second finger; SDL2's reverse-engineered driver handles this correctly. Confirmed: **12-bit X and 12-bit Y per touchpoint** at the input report rate. Cycling '74

Open-source driver validation: `hid-playstation.c` registers a multi-touch input device via `input_mt_init_slots`. Linux mainline driver parses the touchpad fields exactly per the descriptor.

Per-firmware-version differences: No documented changes to touchpad encoding across firmware revisions through 0630 (2025-09-17).

Literature anchor

- Frank, Biedert, Ma, Martinovic, Song, "Touchalytics: On the Applicability of Touchscreen Input as a Behavioral Biometric for Continuous Authentication," *IEEE TIFS* 8(1):136–148 (2013): **30 behavioral features on N=41 users (smartphone), inter-session EER 2–3%, inter-week EER 0–4% median.** Most informative features per relative mutual information: area covered by fingertip, stroke velocity, fingertip pressure.
- Sitová et al., "HMOG: New Behavioral Biometric Features for Continuous Authentication of Smartphone Users," *IEEE TIFS* 11(5), 2016: hand movement + orientation + grasp; multimodal fusion best EER ~7%. arxiv
- Antal et al. 2015 (above); pressure features carry significant discriminative weight, but the DualSense touchpad does not expose a pressure proxy.
- Georgiev, Eberz, Turner, Lovisotto, Martinovic, "Common Evaluation Pitfalls in Touch-Based Authentication Systems," *ASIA CCS* 2022: touch-dynamics studies systematically over-report performance due to evaluation pitfalls; realistic deployment EERs are several percentage points worse than literature claims. **This caveat applies directly to any v2 touchpad feature claim.** Springer

Inter-subject separation-ratio estimate

The DualSense touchpad's 12-bit position resolution at the input-report rate is sufficient to support stroke-velocity, multi-touch coordination, and inter-touch timing features. The two missing Touchalytics features (contact area, pressure) are the highest-mutual-information features in Frank et al. — losing them shifts realistic EER from the 2–3% range toward the 5–10% range. Separation ratio remains comfortably above the 0.5 floor.

The binding constraint is **data volume per session**, not per-event separability. The DualSense touchpad is rarely engaged during competitive gameplay (it is primarily a menu/map navigation surface), so the per-session feature volume is small. This positions touchpad as a **co-signal** that contributes when engaged but cannot be relied on as a primary discriminator.

Adversarial spoof cost

Translator-class hardware (Cronus Zen, XIM, reWASD) typically remaps or ignores the touchpad surface entirely; the Cronus Zen "Expert Controller GamePack" reassigns the touchpad to Select/Back/View/Minus when a non-DualSense controller is connected. **Synthesizing realistic two-finger swipe geometry over time is feasible at the byte level** because the surface is position-only — no physiological constraint is encoded — and there are no known cryptographic protections on the touchpad telemetry. Spoof cost: **moderate** if a translator vendor decides to target the surface; trivial in the byte-injection threat model where the cheat already has firmware-level intercept. This further argues for co-signal status. Cronusmax

Recommended L4 inclusion tier: CO-SIGNAL

Surface 3 — Microphone Array Acoustic Fingerprinting (CONDITIONAL / DROPPED, 15% weight)

Vendor-spec confirmation

The DualSense exposes audio over USB Audio Class 1.0 (UAC1). Per Cristian Ciocaltea's May 2025 patch series ("HID: playstation: Add support for audio jack handling on DualSense," LWN.net 1022523): *"the controller complies with v1.0 of the USB Audio Class spec (UAC1)."* On Linux this surfaces as an ALSA card `alsa_card.usb-Sony_Interactive_Entertainment_DualSense_Wireless_Controller-00` with input and output PCM endpoints. **Mic only available over USB; built-in microphone is unsupported over BT** (PCGamingWiki; BoilingSteam first-look review). Phoronix + 2

The Game Controller Collective Wiki entry describes the controller's microphone path as "Mono? Body Mic with Back Mic sound canceling" — **only a single mono channel is exposed to the host over UAC1, despite the on-controller multi-mic noise-cancelling DSP** (Realtek ALC5524 audio codec, Sony codename "Venom," per psdevwiki). Fandom

This is a critical finding: **the host receives a single mono audio stream, not a multi-mic array.** Most of the room-impulse-response and beam-forming literature requires multi-mic input. The DualSense's on-controller back-mic sound-cancelling DSP destroys the per-mic phase information before the audio reaches the host.

Open-source driver validation: ALSA usb-audio quirk (Ciocaltea 2025) plus the Linux generic UAC1 path. No vendor-specific HID controls for raw per-mic audio access have been reverse-engineered as of this document.

Literature anchor

- Boger-Lombard, Slobodkin, Katz, "Non-Line-of-Sight Passive Acoustic Localization Around Corners," Hebrew University, arXiv 2211.03453 (2022): passive acoustic localization in reverberating rooms using broadband noise correlation across **multiple detectors** — does not transfer to a single mono mic. arxiv
- Gabbrielli et al., "Comparison of Direct Intersection and Sonogram Methods for Acoustic Indoor Localization of Persons," *Sensors* (PMC 8271839, 2021): person localization to ~0.3–0.9 m mean absolute error using room impulse response, **multi-channel.** nih

- Xue, Zhuge, Wang, *Sensors* 26(2):717 (2026): smartphone-as-receiver indoor acoustic positioning using vision-derived priors — single-mic precedent, but requires controlled active anchors emitting near-ultrasonic signals. nih
- US Patent 9,449,613 ("Room identification using acoustic features in a recording") describes machine-learning-based room fingerprinting from "ordinary audio recordings" — proves feasibility of single-mic room-RIR features but raises every privacy issue noted below. uspto

Privacy analysis (the falsification test)

The position to falsify is whether the microphone array can be incorporated as a presence-attestation channel without unacceptable privacy cost. The result: **falsified at the default scope** under at least three jurisdictions:

- **Illinois BIPA (740 ILCS 14):** Voiceprints are an explicitly enumerated biometric identifier. Concrete 2024–2025 litigation anchors: The Lyon Firm
  - *Bell v. Petco Animal Supplies Stores, Inc.*, Case No. 1:22-cv-06455 (N.D. Ill.) — Judge Sunil Harjani granted preliminary approval March 14, 2024, to a **$445,000 settlement covering 445 warehouse workers** who used a Honeywell Vocollect voice order-picking system; per Bloomberg Law: *"An Illinois federal judge granted preliminary approval Thursday to a $445,000 settlement between Petco and 445 warehouse workers who accused the pet supply chain of unlawfully capturing, storing and using their voiceprints through headsets they used to navigate work tasks."* That is $1,000 per class member, the BIPA negligent-violation floor. The Lyon Firm + 3
  - *Parker et al. v. Verizon Communications Inc. et al.*, Case No. 1:24-cv-08436 (N.D. Ill. Eastern Division), filed September 18, 2024 — targets Verizon's Voice ID program; Judge Jorge Alonso ordered the case to arbitration on May 30, 2025. The Lyon Firm IdTechWire
  - *Cruz v. Fireflies.AI Corp.*, No. 3:25-cv-03399 (C.D. Ill.), filed December 18, 2025 — plaintiff Katelin Cruz; targets Fireflies' "Speaker Recognition" feature which generates voiceprints of *non-account-holding meeting participants*. Per the National Law Review / Workplace Privacy Report: *"The plaintiff in Cruz v. Fireflies.AI Corp., No. 3:25-cv-03399 (C.D. Ill.), alleges that she participated in a virtual meeting hosted by an Illinois nonprofit organization that had enabled Fireflies.ai."* The non-account-holding-third-party theory in *Cruz* is directly analogous to a household member whose audio is incidentally captured by a DualSense mic in a tournament-from-home configuration. tl;dv + 3
  - **Statutory damages: $1,000 per negligent violation, $5,000 per intentional/reckless violation, per scan.** Even framing the use as "presence attestation, not voice ID" does not avoid BIPA: the trigger is the *capture and storage* of biometrically-identifiable audio. NatLawReview UMEVO

- **EU GDPR Article 9** prohibits processing biometric data for the purpose of uniquely identifying a natural person without one of the narrow lawful bases. The EU AI Act's biometric-categorization rules further restrict any inference of human characteristics from biometric input.
- **US Federal Wiretap Act and state two-party consent regimes** (CIPA in California, Florida, Pennsylvania, Illinois 720 ILCS 5/14-2): capture of "oral communications" requires consent; tournament terms-of-service consent does not necessarily reach household third parties whose audio is incidentally captured. The Lyon Firm

A narrow non-speech feature subset may survive: running averages of spectral centroid below the speech band, room reverberation time (T60) estimates from impulse responses to controller-emitted speaker challenges, and multi-second-binned envelope statistics. These do not contain phonetic content and do not reconstruct identifiable voiceprints. However: the processing must be **on-device or in a tightly-bounded enclave** to avoid raw audio touching infrastructure where consent regimes attach. The DualSense controller does not provide such an enclave; processing would have to be on the host PC, which means raw audio crosses host-OS boundaries. **There is no published anti-cheat work using even the narrow non-speech feature subset — this would be genuinely novel, but is also a privacy-litigation magnet.**

Adversarial spoof cost

A translator-class device does not pass through controller mic audio by default. Spoofing a static room-fingerprint stream is **trivial at byte level** if the threat model includes firmware-level controller emulation; the audio payload is unauthenticated UAC1 PCM. The forensic value would have to come from *consistency over session* rather than per-frame discriminability.

Recommended L4 inclusion tier: DROPPED at default scope; RESEARCH-ONLY for the narrow non-speech feature subset.

The output for this surface, per the brief, is a feasibility-and-privacy-constraints document, not a feature specification. **Do not implement microphone capture in v2 L4.** If the protocol later requires a passive-presence channel, the camera-based optical-witness path on Surface 4 is strictly preferable on every privacy axis.

Surface 4 — Lightbar Optical-Witness Emission (CO-SIGNAL, 12% weight)

Vendor-spec confirmation

The DualSense lightbar is an RGB LED ring around the touchpad (per `controllers.fandom.com` Sony DualSense entry: "RGB LED (24 bit color)"; not a back-of-controller bar like the DS4). It is host-controllable via output report: Fandom PCGamingWiki

- Linux mainline `hid-playstation.c` registers it as a multicolor LED via `led_classdev_mc` with sub-LEDs for `LED_COLOR_ID_RED`, `LED_COLOR_ID_GREEN`, `LED_COLOR_ID_BLUE` (Roderick Colenbrander patch series, "[PATCH v6 1/4] HID: playstation: add DualSense lightbar support"). Spinics.net
- The `dualsense-controller-python` README documents the output-report bytes 45–47 as `color: ff ff ff` (R, G, B 8-bit each) plus `lightbar_pulse_options` and `led_options`. PyPI PyPI
- Brightness levels exposed: `Bright`, `Mid`, `Dim` (3 levels) per the `LightBrightness` enum in the Game Controller Collective Wiki Data Structures page. Fandom
- Update rate is bounded by the host output-report rate (≈250 Hz over USB at standard polling, ≈1000 Hz over the Edge's 1 kHz USB polling). **Lightbar PWM frequency for the LED hardware itself is not externally specified by Sony**, but downstream RGB-LED PWM controllers in this device class typically run in the kHz range and would be invisible to a 30/60 fps camera.
- **Per-firmware-version differences:** firmware 0630 added multi-device pairing without changing lightbar control encoding.

Literature anchor (visible-light covert channels and screen-camera communication)

- Hao, Zhou, Xing, "COBRA: Color barcode streaming for smartphone systems," ACM MobiSys 2012. ResearchGate ACM Digital Library
- Hu, Gu, Pu, "LightSync: unsynchronized visual communication over screen-camera links," ACM MobiCom 2013. ACM Digital Library
- Li, An, Campbell, Zhou, "HiLight: Hiding bits in pixel translucency changes," ACM Workshop on Visible Light Communication Systems (VLCS) 2014. ACM Digital Library
- Kuo, Pannuto, Hsiao, Dutta, "Luxapose: Indoor positioning with mobile phones and visible light," ACM MobiCom 2014. ACM Digital Library
- Ghiasi, Zúñiga Zamalloa, Langendoen, "A Principled Design for Passive Light Communication," ACM MobiCom 2021. ACM Digital Library

These show that 30–60 fps cameras can reliably decode visible-light symbol streams from screen-class sources at multi-meter ranges; PWM-modulated RGB LED sources at lower modulation rates are even more permissive.

Engineering path for third-party witness

Tournament station camera (referee's phone, e.g., 30 fps 1080p, or a webcam) observes a host-issued lightbar challenge pattern signed with the controller's secret. Practical pattern: 3-color symbol stream at 5–15 symbols/second over a 5-second authentication window, producing 25–75 bits — sufficient for a Merkle-proof challenge response keyed off the controller's session secret. This is comparable in role to the LAN-tower BlueZ witness path established in the BT calibration prerequisite, with the advantage that **the witness is fully passive (camera) rather than requiring an active BlueZ host on the same RF channel.**

Inter-subject separation-ratio estimate

The lightbar is a host-controlled output, not a player-discriminative input. **It carries zero per-player discriminative information by itself.** Its value is as a challenge-response witness channel, not as a feature source.

Adversarial spoof cost

Spoofing the lightbar value at byte level is trivial (translator-class output remap). Spoofing the *physical optical emission* observed by an independent camera that does not connect to the host PC is structurally equivalent to defeating the BT witness path: the cheat would need to either compromise the witness device, add a second emitter at the same spatial location, or maintain real-time MITM between the host and a real controller. **Spoof cost vs. independent witness: high.**

Recommended L4 inclusion tier: CO-SIGNAL (challenge-response witness only).

Surface 5 — Battery Drain Multi-Component Signature (ADVISORY ONLY, 10% weight)

Vendor-spec confirmation

Battery telemetry is exposed via the input report's status byte:

- The battery field is **a 4-bit power level value in the range 0x00–0x0A (11 distinct levels, ~10% per step)** per the neolit123/go-dualsense-battery analysis: *"The battery power level is stored in 4 bits of memory in the input report from the controller. That is a value between 0x00 and 0x0F, or 16 levels in total. However, the actual value is in the range between 0x00 and 0x0A, or 11 levels in total."* GitHub
- A `PowerState` enum (`Discharging`, `Charging`, `Complete`, `AbnormalVoltage`, `AbnormalTemperature`, `ChargingError`) per Game Controller Collective Wiki `Data_Structures` page. Fandom
- `pydualsense` exposes battery state as a high-level `battery.on_change`, `on_lower_than`, `on_charging`, `on_discharging` event API; the underlying field is the 4-bit level above. PyPI
- **There is no exposed instantaneous voltage, current, or drain-rate field over HID.** Telemetry is the bucketed level only. The PMIC is a Dialog DA9087 (per psdevwiki); raw instantaneous power is internal to the PMIC and not surfaced.

- **Update cadence:** battery telemetry updates appear in input reports continuously, but the *value* changes only when the level crosses one of the ~10% bucket boundaries — observed real-world cadence is on the order of multiple minutes per bucket transition under normal load.

Literature anchor

- Yan, Li, Naili, Touati, Wang, "On Inferring Browsing Activity on Smartphones via USB Power Analysis Side-Channel," *IEEE TIFS* (2017): smartphone webpage identification via USB charging-station power side-channel — **>90% accuracy** under controlled conditions, **~2.2% across-70-day-gap** — high specificity but high temporal sensitivity. W&M ScholarWorks
- La Cour et al., "Wireless Charging Power Side-Channel Attacks," arXiv 2105.12266 (2021): wireless-charger power side-channel can fingerprint smartphone activity. ACM Digital Library
- WISERS, "Uncovering User Interactions on Smartphones via Contactless Wireless-Charging Side Channel," IEEE S&P 2023.
- Zhang et al., "A Study on Power Side Channels on Mobile Devices," arXiv 1512.07972 (2015): polls Android battery status files at high rate and finds repeatable per-app power signatures. arXiv

Crucial precondition for all of these: they require power-trace sampling rates of **kilohertz to megahertz**, fed by purpose-built measurement hardware or by polling instantaneous voltage/current battery files exposed by the OS. **The DualSense exposes neither** — only the bucketed 11-level state at multi-minute change cadence.

Inter-subject separation-ratio estimate

At the available telemetry resolution, the signature of "controller held and actively played" vs. "controller resting on desk emitting BT keep-alives" is dominated by absolute drain rate over multi-minute windows. The DualSense's mainboard (SIE CXD9006GG, per psdevwiki) plus haptic VCMs plus adaptive trigger VCMs plus radio constitute a load that varies by 5–10× between idle and active engagement — observable at the 11-bucket resolution within a 30–60 minute session, but with no per-event temporal structure. **This is too coarse for primary or co-signal discrimination on per-round/per-match timescales.** Potentially useful as a session-integrity check (controller has not been swapped for an idle one) over 30+ minute windows — advisory-only capability.

Adversarial spoof cost

Trivial. Battery state in the input report is unauthenticated and can be set to any plausible value by translator-class firmware. Replay attacks are essentially free.

Recommended L4 inclusion tier: ADVISORY ONLY.

Surface 6 — Stick Analog Noise Floor + Hall-Effect Stick Fingerprinting (CO-SIGNAL / ADVISORY, 8% weight)

Vendor-spec confirmation

Per the parsed HID descriptor (nondebug/dualsense, Game Controller Collective Wiki, `hid-playstation.c`):

- Bytes 1–4 of input report 0x01 carry left X, left Y, right X, right Y as **8-bit unsigned integers**, neutral ≈ 0x80 (`Generic Desktop / X, Y, Z, Rz`). **The HID-exposed stick ADC precision is 8 bits.**
- The hardware ADC inside the controller is higher-resolution (the SIE CXD9006GG MCU/DSP has multi-channel ADC capability), but the HID encoding truncates to 8 bits per axis. The "8-bit-vs-16-bit debate" is settled: **at the host HID surface, it is 8 bits, full stop.**
- **DualSense Edge replaceable stick modules:** Edge ships from Sony with **ALPS Alpine potentiometer-based modules** (Sony's official PlayStation Direct stick-module replacement product page; ifixit.com/News/71652). Aftermarket Hall-effect (GuliKit, ZeroStick Pro) and TMR (MODDEDZONE, XP Controllers, Battle Beaver Customs) replacement modules exist and are increasingly common in the competitive scene. PlayStation + 2
- The DualSense Edge **trigger** sensors are Hall-effect from the factory (gamepadtest.app DualSense tester); the **stick** sensors are not. iFixit

Stick noise floor and hand-tremor literature

- Lakie, Vernooij, Osborne, Reynolds, "The resonant component of human physiological hand tremor is altered by slow voluntary movements," *J. Physiology* (PMC 3424765, 2012): physiological hand tremor peaks at **8–12 Hz, with 90% of subjects (N=237 in Lakie 1986 reference cohort) showing peak between 7 and 11 Hz.**
- Allum, Dietz, Freund, "Motor-unit activity responsible for 8- to 12-Hz component of human physiological finger tremor," *J. Neurophysiology* (PubMed 943474): peer-reviewed anchor on the 8–12 Hz physiological tremor band in finger-force tasks.
- "Amplitude changes in the 8-12, 20-25, and 40 Hz oscillations in finger tremor" (PubMed 11018494): three stable neural rhythms; the 20–25 Hz band amplitude is modulated by mechanical-reflex feedback. PubMed

Tremor-amplitude vs. 8-bit ADC analysis: stick travel maps to ~10 mm of physical thumb displacement at the stick top. At 8-bit resolution that is ~0.04 mm per LSB. Physiological tremor at the thumb-on-stick-top pose has displacement amplitudes of order 0.05–0.5 mm peak-to-peak in healthy adults. **The 8-bit HID encoding is at the noise floor — tremor is detectable when the**

player is gripping the stick with intent (high amplitude end), but smaller-amplitude tremor signatures will be quantization-limited. A 16-bit encoding would lift tremor signatures clear of quantization, but is not exposed.

A realistic "controller in hand" vs. "controller on desk" classifier using only stick X/Y at 8-bit can detect held vs. unheld at very high accuracy (held controllers show non-zero high-frequency variance in the 8–12 Hz band; placed controllers show none). **As a binary held/unheld discriminator, the stick noise floor is reliable.** As a per-player fingerprinting feature in an N=50 same-population tournament, the 8-bit floor is too coarse to support strong claims.

Hall-effect stick fingerprinting (Givehchian-style physical-layer analog)

Givehchian et al., "Evaluating Physical-Layer BLE Location Tracking Attacks on Mobile Devices," IEEE S&P 2022: 40–47% of mobile devices in field tests had unique BLE physical-layer fingerprints from manufacturing imperfections in the radio. The follow-up "Practical Obfuscation of BLE Physical-Layer Fingerprints on Mobile Devices," IEEE S&P 2024, demonstrates obfuscation can defeat tracking. **The structural analog for Hall-effect stick modules:** per-unit response curve (X-vs-magnetic-field, Y-vs-magnetic-field, deadzone shape, return-to-center bias) carries manufacturing-tolerance variation that could constitute a hardware fingerprint. ScienceDaily Semantic Scholar

Same-controller-population separability constraint (THIS IS A REPEAT OF THE L8 BT FINDING — IT BITES HERE TOO): N≈50 identical DualSense Edge units in a tournament hall, particularly if all running an aftermarket TMR module batch from the same supplier (very common in the competitive Edge scene per the MODDEDZONE, Battle Beaver, and XP Controllers product pages), will sit closer in Hall-stick response-curve feature space than a heterogeneous public population would. The Givehchian-style attack achieves 40–47% unique-identification on a heterogeneous public population; in a same-model same-batch population the hardware variance budget collapses by an order of magnitude or more. **This surface is therefore marked at the matrix level as session-bound presence attestation only — not cross-session controller identity — until a same-model separability study is run.** Any architectural proposal in v2 that relies on Hall-stick fingerprinting as a long-term controller-identity anchor is superseded by this finding.

Adversarial spoof cost

Stick X/Y at 8 bits is the single most heavily-spoofed surface in the entire console-cheat ecosystem (Cronus Zen "Custom Analog Stick Curve" with 10 configurable points, XIM, reWASD). Synthesizing a plausible 8–12 Hz tremor envelope on top of a synthetic stick value is **straightforward** for any party that has read the physiological-tremor literature; this lowers the bar significantly compared to force-curve synthesis on Surface 1. Spoof cost: **low to moderate**. This pushes stick-noise-floor tremor features toward co-signal, not primary, status. Cronusmax

Recommended L4 inclusion tier:

- **Stick analog noise floor → CO-SIGNAL** (held/unheld binary, plus tremor-band variance as supporting evidence).
- **Hall-effect stick per-unit fingerprinting → ADVISORY ONLY, marked session-bound presence only, blocked from cross-session use until a same-model separability study is run.**

Per-Surface Sensor-Stack Matrix

#	Surface	HID encoding (precision, rate)	Vendor-spec verified	OSS-driver verified	Literature anchor	Inter-subject separation (literature)	Adversa spoof co
1	Adaptive trigger force-curve	8-bit per trigger axis @ ≈1 kHz on Edge USB	✓ (PSDevWiki, GCC Wiki, SensePost, nondebug, pydualsense)	✓ (hid-playstation.c, DualSense-Windows)	✓ (Saevanee 2009; Antal 2015; Wakita 2006; Miyajima 2007; Van Vugt 2013)	EER 1–15% lab→field; ID 70–90% on N=12–276	Very hig public fo curve re demonst
2	Touchpad capacitive	12-bit X/Y, 2-point, no pressure proxy	✓ (PSDevWiki, GCC Wiki, nondebug, Cycling '74)	✓ (hid-playstation.c)	✓ (Frank 2013; Sitová 2016; Antal 2015; Georgiev 2022)	EER 2–4% (Touchalytics best); 5–10% realistic without pressure feature	Moderat level syn feasible)
3	Microphone array	UAC1 mono PCM (single channel, on-controller noise-cancellation DSP)	✓ (psdevwiki, Ciocaltea ALSA quirks, BoilingSteam)	✓ (ALSA usb-audio quirk patches 2025)	Partial — multi-mic literature does not transfer to single mono channel	N/A — falsified at default scope by privacy regime	Low at b level

#	Surface	HID encoding (precision, rate)	Vendor-spec verified	OSS-driver verified	Literature anchor	Inter-subject separation (literature)	Adversa spoof co
4	Lightbar	24-bit RGB at host output rate	✓ (PSDevWiki, GCC Wiki)	✓ (hid-playstation.c lightbar registration)	✓ (COBRA, HiLight, LightSync, Luxapose, Passive Light Comms)	N/A as discriminator; ~25–75 bits/5 s as challenge channel	High vs. indepen camera v trivial vs only
5	Battery drain	4-bit (11-bucket) level, multi-minute change cadence	✓ (GCC Wiki Data_Structures, neolit123 analysis, pydualsense)	✓ (hid-playstation.c battery parsing)	✓ (Yan 2017; WISERS 2023; Zhang 2015)	N/A — telemetry resolution too coarse	Trivial
6	Stick noise floor / Hall fingerprint	8-bit per axis stick X/Y, tremor at 8–12 Hz	✓ (PSDevWiki, GCC Wiki, nondebug, ifixit, gamepadtest)	✓ (hid-playstation.c)	✓ (Lakie 2012; Allum tremor; Givehchian 2022)	Held/unheld near-perfect; per-player marginal at 8-bit	Low-mo (tremor envelope biomech

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Empirical Unknowns (Pre-Corpus Measurements Required Before v2 L4 Architecture Proceeds)

1. **Adaptive trigger force-curve sample-to-sample autocorrelation in same-player same-game-event windows.** The pull biometric literature is on either driving simulators (different actuator) or piano keys (different actuator) or smartphone touchscreens (different muscle group). Required: N=10 players × 100 trigger pulls × 3 game contexts (semi-auto fire, full-auto fire, accelerator hold), measure pull-to-pull intra-player

Mahalanobis distance and player-to-player inter-player Mahalanobis distance. Target: confirm separation ratio ⩾ 1.0 (not 0.5) for primary-discriminator status.

2. **DualSense Edge polling rate stability under host-side load.** The default 1 kHz USB rate is documented but not characterized for jitter under heavy CPU load. Required: jitter histogram at 99th-percentile during a typical Black Ops 7-class game session.
3. **Per-firmware-version trigger telemetry encoding stability.** Firmware 0630 added multi-device pairing; check whether earlier 0500-class and later 0630+ firmwares produce bit-identical trigger byte streams under identical mechanical input.
4. **Hall-effect stick same-model same-batch separability.** The brief explicitly flags this; no literature exists. Required: N=20 stock Edge units (CFI-ZCP1) plus N=20 of one popular aftermarket TMR module batch (e.g., GuliKit), measure deadzone, return-to-center bias, response-curve linearity, X-Y crosstalk per unit. Determine whether unit-level discrimination is feasible in a same-model population.
5. **Lightbar PWM modulation frequency.** Sony does not publish; 30/60 fps cameras must capture stable colors, but whether a faster strobe carrier would interfere with the optical-witness symbol stream is unknown.
6. **Touchpad data volume per typical competitive session.** Without a measurement of how many touchpad events occur in a typical 20-minute Warzone or Apex match, the touchpad-as-co-signal contribution to a fused L4 model is speculative.
7. **Battery-state field cadence under continuous heavy haptic + adaptive-trigger + lightbar load.** Test whether the 11-bucket coarse field changes faster than multi-minute under maximum-load conditions; if so, it might rise from advisory to weak co-signal status.
8. **Whether the ALSA mic-stream sample rate floor (UAC1 typically supports 8/16/32/48 kHz) can be set to a sub-speech-band rate by host-side configuration without provoking the controller firmware to renegotiate or fail.** This determines whether the narrow non-speech RIR feature path on Surface 3 is even technically reachable from a Linux/Windows host.

Same-Controller-Population Separability Constraint (Cross-Surface)

Per the L8 BT lesson, surfaces whose discriminative variance is dominated by inter-unit manufacturing tolerance collapse in a same-model same-batch population. Applying the constraint to each surface:

- **Surface 1 (trigger force-curve):** discriminative variance is dominated by *player biomechanics*, not unit hardware. Constraint does not bite. ✅
- **Surface 2 (touchpad):** variance dominated by player swipe/tap behavior. Constraint does not bite. ✅
- **Surface 3 (microphone):** dropped on privacy grounds; constraint moot. ⏸

- **Surface 4 (lightbar):** challenge-response, no discriminative use. ✅
- **Surface 5 (battery):** advisory only; insufficient resolution for hardware-fingerprint claims regardless of population. ✅
- **Surface 6 (stick Hall fingerprint):** discriminative variance is dominated by *manufacturing tolerance*. **Constraint bites hard. Surface marked session-bound presence only, blocked from cross-session controller identity until same-model separability study is run.** ⛔

Prior-Art Comparison

(a) Anti-cheat literature

- Dorner & Klausner, "If It Looks Like a Rootkit and Deceives Like a Rootkit: A Critical Examination of Kernel-Level Anti-Cheat Systems," ARES 2024 (DOI 10.1145/3664476.3670433): audits BattlEye, Easy Anti-Cheat, FACEIT AC, Vanguard against seven rootkit-behavior metrics (evasion, virtualization, time-of-execution, remote access, information exfiltration, network manipulation, removability). **All four are kernel-level host-OS instrumentation.** None collect any controller-side sensor signal beyond what the game already consumes. Emergent Mind The Moonlight
- RICOCHET Anti-Cheat Season 02, published at https://www.callofduty.com/blog/2026/02/call-of-duty-black-ops-7-ricochet-anti-cheat-season-02 (launched February 5, 2026): introduces TPM 2.0 + Secure Boot + Microsoft Azure Attestation cloud-based PC integrity attestation, plus behavior-based machine-modified-input detection (timing / consistency / response patterns). **No controller sensor fusion.** Insider Gaming + 2
- RICOCHET Anti-Cheat Season 05, published at https://www.callofduty.com/blog/2025/08/call-of-duty-black-ops-6-ricochet-anti-cheat-season-05: kernel-level driver and server-side ML; explicitly states "the RICOCHET Anti-Cheat kernel-level driver does not operate unless Call of Duty is running" — i.e., no controller-side telemetry path. Call of Duty Call of Duty
- Tencent ACE, Riot Vanguard, EAC public documentation (2025–2026 updates): all operate via host-OS drivers and process/memory introspection. None document controller sensor-fusion features.

Confirmation: none of the audited anti-cheat systems collect adaptive-trigger force-curves, touchpad capacitance, controller-mic acoustic fingerprints, lightbar optical proofs, or battery drain signatures. **VAPI is structurally first in this surface set.**

(b) Human-factors and biometric-research literature

Already enumerated per surface. Key anchors: Frank Touchalytics 2013 (touchpad), Wakita /

Miyajima 2005–2007 (pedal force), Van Vugt 2013 (key-velocity), Saevanee 2009 (press pressure), Lakie 2012 / Allum (physiological tremor), Boger-Lombard 2022 (passive acoustic localization), COBRA / HiLight / Luxapose (visible-light covert channels), Yan 2017 / WISERS 2023 (power side channels), Givehchian 2022/2024 (BLE physical-layer fingerprinting).

(c) DualSense reverse-engineering work

nondebug DualSense Explorer (github.com/nondebug/dualsense), `dualsensectl`, the SensePost reverse-engineering effort (sensepost.com/blog/2020/dualsense-reverse-engineering/), psdevwiki.com/ps5/DualSense, BlueRetro project documentation, `pydualsense` (flok), `dualsense-controller-python` (yesbotics), DualSense-Windows (Ohjurot), Game Controller Collective Wiki (controllers.fandom.com), Linux mainline `drivers/hid/hid-playstation.c` (Roderick Colenbrander, Linux 5.12, April 2021), the ALSA usb-audio Sony DualSense quirk patch series (Cristian Ciocaltea, Collabora, May 2025). PyPI

This corpus collectively constitutes the verification class the BT-CALIB-LESSON-001 application rule requires; it is sufficient to verify Surfaces 1, 2, 4, 5, 6 and partially sufficient for Surface 3 (mono mic stream is verified; multi-mic per-channel access is not exposed).

Recommendations (Staged, Decision-Ready)

Stage A (immediate, before v2 L4 architecture spec is drafted):

1. Lock in Surface 1 (adaptive trigger force-curve) as the v2 L4 primary discriminator. Allocate engineering time to a 1 kHz USB capture pipeline that records `Rx` and `Ry` bytes at the polling-rate floor.
2. Run Empirical Unknown #1 (intra-player vs. inter-player Mahalanobis distance on trigger pulls) on N⩾10 players. **Decision threshold:** if separation ratio ⩾ 1.0, force-curve graduates to primary discriminator; if 0.5–1.0, demote to co-signal; if < 0.5, drop and revisit Surface 6 fusion strategy.
3. Run Empirical Unknown #4 (Hall-effect stick same-model separability) on N=20 stock + N=20 batched-aftermarket Edge units. **Decision threshold:** if unit-level discrimination > 50% rank-1 on heterogeneous models AND > 20% on same-model same-batch, Hall fingerprint moves to co-signal status; otherwise it remains advisory-only and session-bound.

Stage B (after Stage A measurements land):

4. Implement adaptive-trigger challenge-response (Surface 1 ↔ L6 coupling): host issues a `Slope` or `Vibration` resistance profile, observes the player's force-vs-time response over the next 200–500 ms, validates against an enrollment model. This converts the trigger surface from a passive feature source into an active anti-cheat challenge channel.

5. Implement lightbar optical-witness path (Surface 4) as an alternative to the LAN-tower BlueZ witness in tournament configurations where the BlueZ host is impractical. Target symbol rate 5–15 sym/s over a 5-second window.
6. Add touchpad swipe-velocity / multi-touch-coordination features (Surface 2) as fused co-signal weighted by per-session touchpad event count.

Stage C (deprioritized; do not staff until Stage A and B ship):

7. Battery drain (Surface 5) implemented at session-integrity level only: flag suspicious "controller swapped to idle one" events at the 30+ minute window scale.
8. Microphone-array research path (Surface 3): keep on the research radar only if a tightly-scoped non-speech-band on-device feature pipeline becomes architecturally feasible. **Do not collect raw audio on host PCs in production.** Privacy counsel sign-off is a hard prerequisite before any Stage-C work begins on this surface.

Thresholds that change these recommendations:

- If Sony ships a firmware that exposes a 16-bit trigger or stick encoding, Surface 6 noise-floor tremor features rise from co-signal to candidate-primary.
- If a public PoC of a force-curve replay attack appears (e.g., a Cronus-class device shipping a "biomechanical pull synthesis" gamepack), Surface 1 demotes from primary to co-signal and the protocol must add a second primary discriminator on a different surface.
- If a US federal preemption of BIPA voiceprint claims passes Congress, Surface 3 may re-enter feasibility scope — but not before.

Caveats

1. **Privacy:** Surface 3 (microphone array) is conditionally dropped on BIPA / GDPR / wiretap-law grounds. The current case-law anchors are *Bell v. Petco* ($445,000 settlement, March 2024), *Parker v. Verizon* (filed September 2024, sent to arbitration May 2025), and *Cruz v. Fireflies.AI* (filed December 2025) — these establish that voiceprint capture without written consent is actively-litigated BIPA territory in 2024–2025. Any future v2 architectural attempt to revisit this surface must re-run the privacy analysis with current case law. Producing a non-speech-band feature pipeline on the host PC, where raw audio crosses host-OS boundaries, is itself a litigation-risk surface even if the features stored downstream are not voiceprints.
2. **Vendor-firmware-version drift:** Trigger encoding, lightbar control encoding, and battery field encoding are unchanged through firmware 0630 (2025-09-17). Any future firmware

that changes these would require re-verification; the v2 architecture should pin a firmware-version assumption and revalidate at each Sony firmware release.

3. **DualSense Edge stick module heterogeneity:** Edge ships from Sony with potentiometer modules; the competitive scene increasingly uses aftermarket Hall-effect / TMR replacements (MODDEDZONE, GuliKit, ZeroStick Pro, Battle Beaver, XP Controllers). Any feature claim against "DualSense Edge stick" must account for which module class the player is on. The HID encoding does not distinguish — module type is not advertised. This is a measurement gap.
4. **Polling-rate dependence:** The 1 kHz USB polling rate on the Edge is documented but is a default that DSX, reWASD, and similar host-side software can change. Trigger force-curve features that depend on dense temporal sampling must validate at the lower 250 Hz floor for fault tolerance.
5. **Translator-class threat model evolution:** RICOCHET Season 02 explicitly states cheat-device vendors "are designed to hide, adapt, and change configurations to avoid simple detection" — VAPI's force-curve discriminator is currently outside the threat model these vendors target, but if VAPI ships and publishes, a force-curve replay capability could appear in commercial cheat hardware within a typical 6–18 month anti-cheat-vs-cheat cycle. Plan for this. Call of Duty
6. **The 8-bit stick HID encoding is at the floor for tremor detection.** Any v2 stick-tremor feature claim is single-bit-of-margin against quantization. A sensor-fusion approach combining stick noise floor with IMU micro-tremor is strongly advisable rather than relying on stick noise floor in isolation.
7. **The same-controller-population separability constraint applies to L8 BT (already documented) and to Hall-effect stick fingerprinting (Surface 6 in this document). Any future surface added to v2 whose discriminative variance is dominated by manufacturing tolerance must explicitly assess this constraint at proposal time, not at integration time.**
8. **Replication caveat on Enev 2016 driver-fingerprinting claim:** the headline "100% identification from brake pedal alone (N=15)" did not reproduce in Gahr et al.'s broader 2018 study. Trigger force-curve performance projections that lean on Enev as their anchor are optimistic; Miyajima 2007 (76.8% on N=276, real-world city driving) is the more defensible literature anchor and is the one used in the matrix. arXiv